

AFL Prediction Project — Data Dictionary & Target Definitions

Week 3 · Day 1 — Data Foundations contract for all downstream models & the chat agent

1. Source Tables

Table	Grain (1 row =)	Key columns
players_info	One player, career-level bio	id
round_by_round	One player's performance in one match	player_id, year, round, team
seasonal	One player's totals for one team-season (split by finals)	player_id, year, team, is_finals
team_matches	One team's view of one match (2 rows/match)	match_date, venue, team_name, opponent

Coverage: 1983–2025 (43 seasons), 64 venues, ~2,850 players. Real match grain must be reconstructed as (date, venue, sorted team pair) — the raw 'id' column in team_matches is a row id, not a shared match id.

2. Key Data-Quality Fixes Applied

Issue	Fix
Western Bulldogs stored as 'W. Bulldogs' (subject team) vs 'Western Bulldogs' (opponent)	Canonical team-name map applied before any join
Leading/trailing whitespace & tabs in team names	Stripped on load
seasonal.player_id has 'ID_' string prefix on 10 rows	Prefix stripped, cast to Int64
~0.8% of round_by_round player_ids have no bio record	Left as unmatched; flagged, not dropped
Small number of exact duplicate rows in all 3 stat tables	Dropped via drop_duplicates()
Many advanced stats (contested poss., hit-outs, etc.) NaN in early seasons	Left as NaN — reflects stats not tracked that era, not true zeros

3. Prediction Targets

Target	Level	Definition / Formula
match_result	1 / match	3-class: Home Win / Away Win / Draw, from match_margin sign
match_margin	1 / match	home_score – away_score (secondary regression target)
top_disposals	1 / player-match	kicks + handballs (raw column: disposals)
top_goalkicker	1 / player-match	goals (raw column)
fantasy_score	1 / player-match	3·kicks + 2·handballs + 3·marks + 4·tackles + 1·hit_outs + 1·FF – 3·FA (validated $r \approx 0.98$ vs. provided fantasy_points)

Match winner uses classification (not pure margin regression) because the practical question — for both the ML system and the chat agent — is a discrete 'who wins' outcome. Margin regression is retained as a secondary target since it can always be thresholded into W/L/D.

4. Feature Table (versioned, see notebook Task 4)

Feature	Window	Leakage-safe?
win_rate_last{3,5,10}, avg_score_for/against_last5	Rolling N, shift(1)	Yes — excludes the match being predicted
win_streak, rest_days	Expanding / diff, shift(1)	Yes
ladder_pts_cum_pre, ladder_rank_pre	Season-to-date, shift(1)	Yes
h2h_win_rate_prior, venue_win_rate_prior	Expanding, shift(1), grouped	Yes
disposals/goals/fantasy_avg_last{3,5,10} (player)	Rolling N, shift(1)	Yes

5. Train / Hold-Out Split

Strict time-based split: train on all seasons before the holdout year(s); hold out the most recent season(s) entirely. A random split would leak future information — form/ladder features and team strength are serially correlated within a season, so random shuffling lets the model see a team's later-season trajectory while predicting its earlier matches. Reusable split function: time_based_split(df, holdout_seasons=1).

6. Realistic Accuracy Ceiling

Published AFL prediction work and betting-market benchmarks generally place match-winner classifiers around 65–72% accuracy with solid pre-match features, only modestly above the naive home-team baseline (~57–60% in this data). Player output prediction is noisier still. Accuracy near 90%+ on a genuine holdout is a red flag for leakage, not a breakthrough — sports outcomes retain a large irreducible random component (injuries, umpiring, weather) that no amount of feature engineering removes.